

Tesla Stock Price Prediction

Using Deep Learning: SimpleRNN vs LSTM

Field	Details
Project Type	End-to-End Deep Learning — Time Series Forecasting
Dataset	TSLA.csv (2,416 rows, June 2010 – Feb 2020)
Target Variable	Close Price (USD)
Models Built	SimpleRNN, LSTM
Framework	TensorFlow 2.x / Keras
Best Model	LSTM ($R^2 = 0.836$, RMSE = \$30.01)
Forecast Horizons	1 Day, 5 Days, 10 Days
Author	Vijay CyRaksha Siliguri

1. Executive Summary

This report presents a comprehensive deep learning system for predicting Tesla (TSLA) stock closing prices. Two recurrent neural network architectures — SimpleRNN and LSTM — were built, trained, evaluated, and compared. The LSTM model achieved an R^2 of 0.836, significantly outperforming SimpleRNN ($R^2 = 0.602$), demonstrating the superiority of gated memory units for long-term financial time series.

2. Business Problem Statement

Stock price prediction is a critical challenge in quantitative finance. Accurate forecasting enables better trading decisions, portfolio management, and risk mitigation. Traditional statistical models (ARIMA, GARCH) struggle with the non-linear, non-stationary nature of stock data. Deep learning models, particularly LSTMs, can learn complex temporal patterns from historical price sequences.

Business Use Cases

- Automated Trading: Generate buy/sell signals for algorithmic trading systems
- Portfolio Optimization: Dynamically rebalance holdings based on price forecasts
- Risk Management: Forecast value-at-risk (VaR) using predicted price distributions
- Financial Forecasting: Earnings and revenue correlation with stock movement
- Investment Planning: Long-term trend analysis for institutional investors
- Research: Benchmark deep learning architectures for time series

3. Data Understanding

Column	Type	Description	Example
Date	Datetime	Trading session date	2010-06-29
Open	Float	Opening price (USD)	19.00
High	Float	Daily high price (USD)	25.00
Low	Float	Daily low price (USD)	17.54
Close	Float	Closing price — TARGET (USD)	23.89
Adj Close	Float	Split/dividend adjusted close	23.89
Volume	Integer	Shares traded	18,766,300

Descriptive Statistics

Statistic	Open	High	Low	Close	Volume
-----------	------	------	-----	-------	--------

Count	2,317	2,317	2,317	2,317	2,317
Mean	\$68.82	\$70.85	\$66.74	\$68.82	21.4M
Std	\$105.42	\$108.44	\$102.16	\$105.42	23.2M
Min	\$14.98	\$15.80	\$14.25	\$15.01	0.8M
Max	\$786.14	\$968.99	\$676.00	\$780.00	304M

4. Data Cleaning

- Missing Values: Zero missing values found across all columns
- Duplicate Records: Zero duplicate rows after deduplication
- Date Conversion: 'Date' column parsed to datetime and set as index
- Sorting: Index sorted chronologically to maintain time order
- Missing Value Strategy: Forward fill applied (carries last known price)

Missing Value Strategy Comparison

Method	Description	Pros	Cons	Used?
Forward Fill	Use previous value	Preserves trend	Propagates errors	■ YES
Backward Fill	Use next value	Future-aware	Data leakage risk	■ No
Interpolation	Linear/spline	Smooth output	Computationally heavy	■ No

5. Feature Engineering

Feature	Formula	Window	Business Significance
Daily Return	$(Close_t - Close_{t-1}) / Close_{t-1}$	1 Day	Momentum / Trend Direction
MA_20	Rolling mean of Close	20 Days	Short-term trend signal
MA_50	Rolling mean of Close	50 Days	Medium-term trend signal
MA_100	Rolling mean of Close	100 Days	Long-term trend / support
Volatility	Rolling std of Daily Return	20 Days	Risk measurement

6. Deep Learning Models

SimpleRNN Architecture

Layer	Units	Output Shape	Parameters
SimpleRNN	64	(None, 60, 64)	4,224
Dropout	0.2	(None, 60, 64)	0
SimpleRNN	32	(None, 32)	3,104
Dropout	0.2	(None, 32)	0
Dense	32	(None, 32)	1,056
Dense (Output)	1	(None, 1)	33

TOTAL			8,417
-------	--	--	-------

LSTM Architecture

Layer	Units	Output Shape	Parameters
LSTM	64	(None, 60, 64)	16,896
Dropout	0.2	(None, 60, 64)	0
LSTM	32	(None, 32)	12,416
Dropout	0.2	(None, 32)	0
Dense	32	(None, 32)	1,056
Dense (Output)	1	(None, 1)	33
TOTAL			30,401

Training Configuration

Parameter	Value	Rationale
Optimizer	Adam (lr=0.001)	Adaptive learning rate, fast convergence
Loss Function	Mean Squared Error	Penalizes large errors in price regression
Window Size	60 days	~3 months of trading history
Train/Test Split	80% / 20%	Standard for time series evaluation
Batch Size	32	Balanced gradient estimation
Max Epochs	80	EarlyStopping prevents overfitting
EarlyStopping	Patience=10	Restores best weights automatically

7. Model Evaluation Results

Metric	SimpleRNN	LSTM	Winner	Improvement
MSE	2,181.91	900.73	■ LSTM	-58.7%
RMSE (\$)	46.71	30.01	■ LSTM	-35.7%
MAE (\$)	28.02	20.77	■ LSTM	-25.9%
MAPE (%)	7.95%	6.56%	■ LSTM	-17.5%
R ² Score	0.6021	0.8357	■ LSTM	+38.8%

Metric Interpretation

- MSE (Mean Squared Error): Penalizes large errors heavily. LSTM is 58.7% better.
- RMSE (Root MSE): In price dollars — LSTM's \$30.01 vs RNN's \$46.71.
- MAE (Mean Absolute Error): Average |error| — LSTM predicts ~\$20.77 off on average.
- MAPE: Percentage error — LSTM achieves 6.56% error rate vs RNN's 7.95%.
- R² Score: Variance explained — LSTM explains 83.6% vs SimpleRNN's 60.2%.

8. Forecasting Results (LSTM)

Using recursive forecasting from the last 60 days of data (last known close: \$780.00):

Horizon	Day 1	Day 2	Day 3	Day 5	Day 10
Price (\$)	\$544.91	\$552.92	\$557.02	\$558.27	\$546.01
Change vs Last	-30.1%	-29.1%	-28.6%	-28.4%	-30.0%

Note: The large negative change reflects the model predicting a reversion from the extreme rally in early 2020 (data ends Feb 3, 2020 at \$780). The model learned from historical patterns where such spikes are followed by corrections. Recursive forecasting accuracy naturally degrades over longer horizons.

9. Model Comparison & Conclusion

The LSTM model decisively outperforms SimpleRNN across all metrics. The key reasons are:

- **Vanishing Gradient Problem:** SimpleRNN suffers from vanishing gradients on sequences >20 steps, losing early context. LSTM's cell state sidesteps this.
- **Gating Mechanism:** LSTM's forget, input, and output gates selectively retain/discard information, crucial for multi-year stock patterns.
- **Long-term Memory:** With a 60-day window, LSTM effectively learns quarterly cycles. SimpleRNN degrades significantly beyond ~20 steps.
- **Parameter Efficiency:** Despite having 3.6x more parameters (30K vs 8K), LSTM's architecture makes each parameter more impactful.

10. Future Scope & Advanced Improvements

Approach	Description	Expected Improvement
GRU	Gated Recurrent Unit — simpler than LSTM	Comparable to LSTM, faster training
Transformer	Self-attention for long sequences	State-of-the-art accuracy
Sentiment Analysis	Twitter/Reddit NLP signals	+5-15% accuracy boost
Macro Features	CPI, Fed rates, oil prices	Better regime detection
Multi-Stock	Correlation across S&P 500	Improved generalization
Ensemble	LSTM + GRU + XGBoost voting	Reduces prediction variance
Hyperparameter Opt.	Optuna/Keras Tuner	Automated architecture search

■ ■ Disclaimer: This project is for educational and research purposes only. Not financial advice.